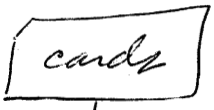
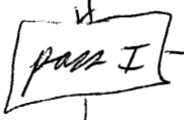
AEC STRETCH Conference Dec 7, 1960 - NYC

(S. L. Prosser) STRAP II

organization



(Tape, Blocked tape, disk later)



→ tape, disk (in disk track sizes)

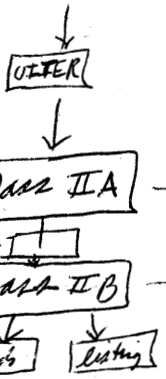
① Expanded Instr, Unit
② Name file

In memory - symbol table

all modes are dittemble ie all data

note: in STRAP I - Card = a card block (a blank in col. I, * in col 1 continuation cards are not in STRAP I.)

StrapII can use any no. of cards; can use 128 non blank characters to a statement name



"UTER" program

assigns location counters — finds "multi-" "undefined" "circularly defined" symbols.

creates binary output + listing also punch symbol table

utilizes this symbol table.

IQS
CC
preset character set, AG
std expanded char set, A8
printer code P8

character sets input will use

STRAP I: REM, SEM pseudo ops in STRAP II (suppress + restore errors)

STRAP II: creates message + holds to end page not as flags,

- four flags which can be suppressed

Time clock reading for ident.

attendees: IBM Hercutz, Groner, Mottram, Mansfield, Carter, Chan, Prosser
R.H. Carter, Nash (IBM-UK), Brannin, Hardy, Kolsky

LASL: C. Folkner, E. Voorhees, E. Kinney, Carter, Jordan, G. Lewis

LRL: G. Boer, Bruijnes, Tiede, Fernbach

AWRE: Jones, Glennie, Armstrong,

TAIL: (similar to HED in 709) have established levels

TAIL: (3) PROSSER

now has to be specified exactly -- could get unintentionally in STRAP I.

UNDEF: Fall at end of program

END: is needed in STRAP II not optional

First version of STRAP II does not depend on MCP

Q: what about "load + go"? - an exit to do this is planned.
some types of errors will prevent going.

~10K words on disk for STRAP II + assoc. progs. (not MCP)

SLCR: "Set Loc Ctr, Relative" - a new pseudo op

Multiple assemblies? - not as yet

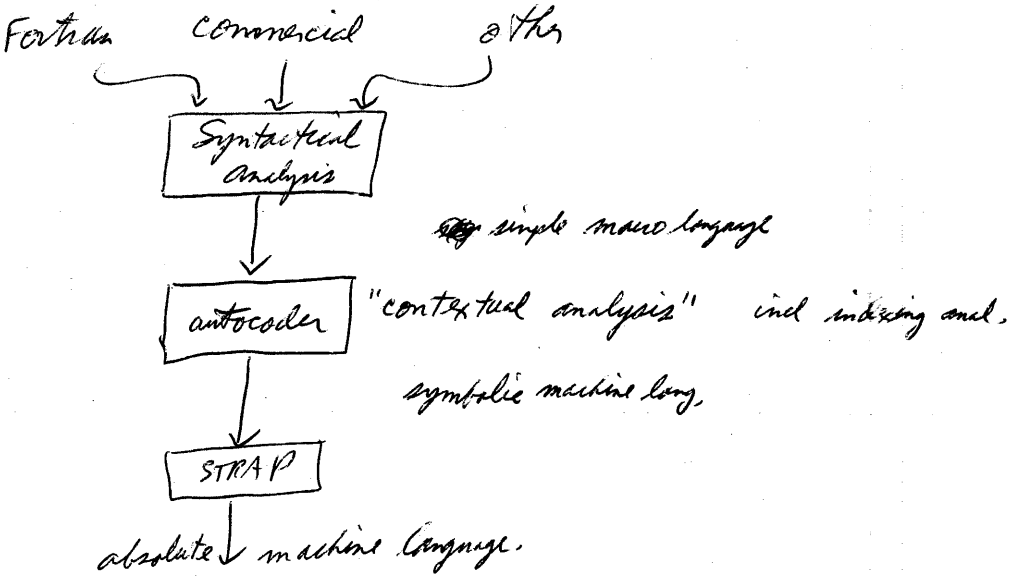
(G. Grove) Autocoder & Fortran

Fortran -- a version of Fortran II incl reloading options slanting toward previous Fortran progs.

autocoder -- an "open" language, macro generator.

eg pseudo ops → calling seqs, plugged in, etc.

– small no. of corrections -- just reinsert



Autocoder: -- a high degree of subroutinizing -- makes easy to change & add to.

Op, P_0, P_1 ... P_n

(1) simplest gives single string of ops out with P's plugged in

(2) conditional strings
descriptor -- generator: output depends on values of P -- decisions dependent on data form, etc, -- conditionality.

(3) establish by data implications

Time schedule:

- planning going on
- part of autocoder done & being debugged
- descriptor language not decided on done

will take to end of '61 to code & Test Fortran, autocoder

- Exact definitions of Fortran? (e.g. in meta language used for algol)
  some translations are done even when not "legal" (e.g. leaving out wpy signs on 709)
- some machine oriented EQUIVALENCE statements setting fixed & fl. data

---

LRL would prefer enrichment of FORTRAN -- eg other modes of arith, buffering of I/O etc. not make 2-way compatible.

IBM - would like to know as early as possible,

Share Organization? Something like SHARE? - smaller not so committee.

⇒ IBM will set up rules for users group -- a proposal before Jan 1, '61

---

question of Translator ... Livermore would do if possible
IBM is interested not committing themselves.

Should it be on '90 or STRETCH?

(G. Cato) COLASL Programs,

- expanded character typewriter, - 3 levels, -- keypunch.
  133 characters? 3 letters per key.

"natural" notation -- ie. textbook or report language

(limited to 12 half line shift up)

A. letters:

simple: $b \; \exists \; \xi \; \Omega$

Super-printed $\vec{F} \; \dot{o} \; \mathring{A}_1 \; \tilde{\epsilon}$ using backspace

over-printed (a) $\boxed{\theta}$

Combinations (π) $\dot{\div} p$ $\boxed{\ddot{\lambda}}$ 2 backspaces,

B. Variable names

simple a $\bar{F}$ $\ddot{a}$ (11)

Prefixed Δr $\gamma \Psi$ ΔW ∂u

scripted F_{max} W_{temp} $\bar{N}_{N23}$ $\alpha_{\bar{A}b}$

C. Control Names:

p #32 TempCal sqrt32a Fabs,

D Constants 3 60.4 4π 9.87×10^{-5} $\frac{3\pi^2}{e}$

II Equations:

A. algebraic → produces code $y = ax^3 + \beta x + c.$ (.111 end of line period plus 2 blanks)

B. parametric $R \equiv 3.629.$ (identity)

→ previous data reservations

III - Program Control Categories

A loop (1st increment last)

1. Forward $i = 1, 2, \ldots, I_{max}$ Through #8 BS

or thru α code $(j = -3, 0, \ldots J)$

2. Reverse $(m = 10, 10 - \Delta m, \ldots, 0)$ from β

3. Fixed: $\begin{cases} y_{i,j} = a_i + b_j \\ z_{i,j} = a_j - b \ldots \\ x_{i,j} = . \end{cases}$

(Some tried to use prepositions not verbs)

for $(i = 1, 2, \ldots 7), (j = J_{max}, J_{max} - \Delta j, \ldots, 0)$

B. Unconditional branching

1. go to #3A (means same as "from #3A")

2. from α can add "proceed from α" (proceed is ignored)

C. Conditional branching

If $(x^2 < y^2 < 4)$ then go to #3B

antecedent consequence

If ( ) and ( ) Then ...

↑ same as "then if"

D. STOP

IV. Array specification

Range: 0 to 3, 9 to 10 for X

Ranges: -1 to 10, -5 to -6, -5 to I_{max} for a, F, Apoz

V Hand Coding

A. attach (reserves for hand coding use)

B. If (X>3) then BBZ, L) asect ← set a bit

VI. Functions (single valued output) & routines (multi-valued output)

A. Specification.

function: λ(a, b, c)

B. Definition.

λ. T = a + 3.7674 y^2 − z

etc.

and end function.

$f^3 g$ means g is dummy function with 3 arguments

VII Comments. (can type in red -- ignored

will add Σ, Π, + some I/O later

2nd Day Thurs. Dec 8, 1960

(Jack Hira) Master Control Program

5 main features

1. automatic operating system
2. system supervisor
3. Parallel multiple I/O Spool
4. Simultaneous I/O assignment
5. Hardware "extensions"

1. auto loading of sequential PP jobs,
2. gives instrs. to operator "mount Reel #n on drive m" etc,
3. accepts instrs. from op

1. Handles binding: compilego, STRAP

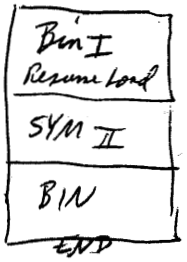
eg. compiles STRAP STRAP → Loader → PP

compiles FORTRAN FORTRAN → SMAC → STRAP → Loader → PP
(SMAC: autocoder)

q. absolutes loader? how to keep from destroying others? - STRAP gives limit words

→ q: making changes to previously compiled program & go,
eg. small changes in big problem & run.

→ This is not now in system,



eg. FORTRAN monitor chain link - Transfer of info from one to other

FBM will look in

Handler requests for LIB and Sys material,

q: is there one dictionary for LIB routines? is there ~~failing~~,
arranged alphabetically – but letters refer to types

A
C
M

form in LIB doesn't matter -- who give form? LIB card specifies?

→ one of main reasons for ~~file protect on tapes~~

- MCP is that there are no hardware file protect on I/O,

control commands are checked in advance – statically at ~~start~~

Multi File I/O – SPOOL

1, parallel processing: I/O, compute
2. automatic buffering
   Tape minimization,
   Multiprog. -- limited – emergency problems given priority, (if it can't use much disk)
   Save & Load

Symbolic I/O assignment

1, PP indep. of I/O configuration –
2. System protection I/O & core
3. Multiprog. compatibility – (later)
4. Tape reel label
5. Restart (later)

"Hardware Extension" — pseudo commands

1. Interrupt cntrl
2. pseudo disable

(Security) SPOOL

Read, Punch, Print etc given symbolically

Input comes from SPOOL

gives MCP chance to analyze in advance — give operator chance to mount reels, etc

can be (1) overlapped
(2) not overlapped

pseudo op: $SCR read card

alternate buffers in mem;

no attempt made to change order of jobs coming in,
i.e. to reject large request & go to next

assume: 1401 C3 (4K Mem)

(Bill Wood) Experience with large program

Monte Carlo – Random walk (Markov chain) Problem

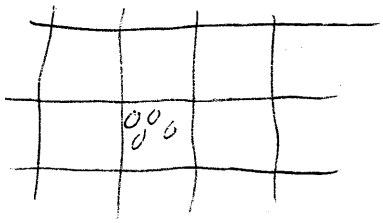
To calculate Equation of State using classical statistical mechanics.

Hard sphere molecules,

10^{21} molecules in 1 gram of water.

use 32 or 256 in 704 cases.

2048 . on STRETCH



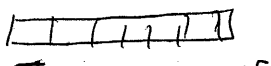
use periodic boundary condition

Ising lattice -- has been solved in some cases - Onsager

- move molecules one by one & check whether overlap has occured.
- calc distances by Pythag. Theorem.

$r_{ij}^2(t)$ for all $\frac{N(N-1)}{2}$ cases.

calculate Eq of state from counts averages



78,000 steps in 10 min on 704.

– in some cases run for 40 hrs. -- worst case, 4 hrs usual.

calc – arranged to never apply

$$r^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$$

use "coarse grained" subdivision - 4096. 2^{12} of phase space

store table of ΔX^2

up to 18 neighbor to any one molecule.

- tables for each molecule.

- random = Fubinacci sequences

each no. = sum of preceding two

use every other one, in sequence

- Time improvement ?? 3+ minutes to do 78,000. steps (10 min on 704)

expected: 20 to 30 speed increase -- 3X only.

(24 μs on 704 vs 1 μs for int ops.)

Flow diag: $\{E_1 + I_1 \{E_2 + C_1 G_1 + E_3\} + C_2 Q + \text{average}\}$

E_1 pull out 5 bits → IT (molecule being displaced)

$X_{IT} \rightarrow X_{IT} + R_X(-\delta, \rho)$

↳ random no.

E_2

C_1 condition G_1: $r^2(IT, J) < \sigma^2$

E_3 count into counters.

C_2 a diffusion test

Q "refresh" neighbor table.

(T, Apple) Simulators

704 simulator on STRETCH will only be ~ 2X 704

- not practical to do 7090 -

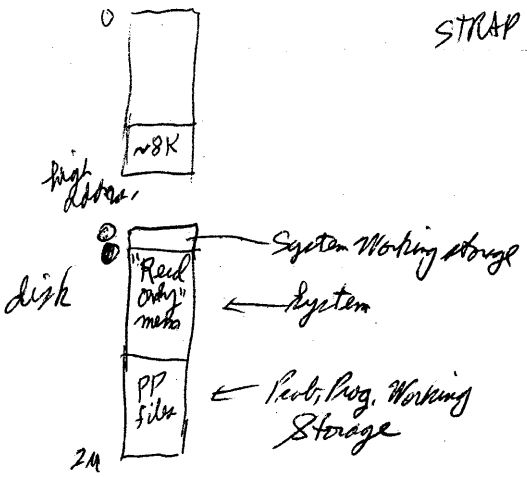
→ Livermore still wants to do it

Load, Dump, Trace Progs.

- each have worked to some extent -- loader is very sensitive to machine state

→ plan to have operator comparing dumps -- will be done soon

(g. groner) Memory allocation



STRAP ≈ 12K but tables are open-ended

average symbol takes ~ 2 words

Livermore Question: what is in 8K ? how much can one call in when he wants it, Spool in -- etc.

Revised Agenda - Stretch Conference
December 7-9, 1960
Time-Life Building
New York City

Wednesday, December 7

9:45 A. M.	STRAP II - - S. L. Prosser
1:15 P. M.	FORTRAN - Autocoder - - G. A. Grover
	COLASL - - LASL

Thursday, December 8

9:45 A. M.	MCP (including SPOOL) -- G. R. Hira, J. L. Garrity, J. M. St. Germaine
1:15 P. M.	Large problem programs -- W. Wood (LASL)
	7090 Simulation -- C. T. Apple
	Disk Simulation -- N. Hardy
	I/O & Editing -- LASL
	System of Present machine usage and state of machine -- C. T. Apple

Friday, December 9

9:45 A. M.	Training & experience -- T. C. Chen & LASL
	Programming systems conventions -- G. A. Grover
	LASL plans and operator training -- LASL
1:15 P. M.	Plans for future meetings
	IVY -- B. Carlson (LASL)
	Programming STRETCH for news optimal performance - - T. C. Chan
	Math subroutines - LASL

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moved to Fri am, Plans for future meetings

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Thurs/pm

p.m. Large problem programs -- LASL (W. Wood)
Observations re parallel file -- (tentative)
Programming STRETCH for near optimal performance -- T. C. Chen
Math subroutines -- LASL

SGC/img
11-30-60